



EMPIRICAL ARTICLE

The effects of maternal input on language in the absence of genetic confounds: Vocabulary development in internationally adopted children

Joseph R. Coffey | Carissa L. Shafto | Joy C. Geren | Jesse Snedeker

Department of Psychology, Harvard University, Cambridge, Massachusetts, USA

Correspondence

Joseph R. Coffey, Department of Psychology, Harvard University, 33 Kirkland St., William James Hall 1080, Cambridge, MA 02138, USA.
Email: jrcoffey@g.harvard.edu

Funding information

National Science Foundation, Grant/Award Number: 0418423

Abstract

Previous studies have found correlations between parent input and child language outcomes, providing *prima facie* evidence for a causal relation. However, this could also reflect the effects of shared genes. The present study removed this genetic confound by measuring English vocabulary growth in 29 preschool-aged children (21 girls) aged 31–73 months and 17 infants (all girls) aged 15–32 months adopted from China and Eastern Europe and comparing it to speech produced by their adoptive mothers. Vocabulary growth in both groups was correlated with maternal input features; in infants with mean-length of maternal utterance, and in preschoolers with both mean-length of utterance and lexical diversity. Thus, input effects on language outcomes persist even in the absence of genetic confounds.

Parents excitedly await their child's first words, often talking and reading to their infants hoping to hurry the process along. The underlying belief, that parents' input matters, is supported by a rich body of research demonstrating that vocabulary growth is correlated with child-directed speech (Brent & Siskind, 2001; Huttenlocher et al., 1991; Newman et al., 2016; Newport et al., 1977; Pancsofar & Vernon-Feagans, 2006; Rowe, 2012; Shneidman et al., 2013). This pattern is both highly generalizable and highly replicable, having been observed across a wide range of cross-linguistic and socioeconomic contexts (Hart & Risley, 1995; Hirsh-Pasek et al., 2015; Hoareau et al., 2019; Hoff, 2003; Hurtado et al., 2008; Huttenlocher et al., 2010; Mastin & Vogt, 2016; Pan et al., 2005; Ramírez-Esparza et al., 2014, 2017; Romeo et al., 2018; Rowe, 2008; Rowe et al., 2017; Shneidman & Goldin-Meadow, 2012; Song et al., 2013; Weber et al., 2017; Weisleder & Fernald, 2013). Many characteristics of child-directed speech have been found to be related to children's vocabulary, such as measures of quantity (e.g., how often children are spoken to) and quality (e.g.,

how diverse or complex speech is). This body of research strongly suggests that input plays a causal role in children's language acquisition: we will refer to this proposal as the *input hypothesis*.

There is, however, a logical alternative to the *input hypothesis*: the correlation between parental input and early word learning could be due to some third factor that affects both input and outcomes. A strong candidate is the shared genes that affect both the parents' behavior and the traits that they pass down to their child, resulting in a passive gene–environment correlation (Plomin et al., 1977; Scarr & McCartney, 1983). Genetic variability contributes to variation in almost all human traits, including language (Polderman et al., 2015). If genetic variability related to linguistic behavior is correlated with both parental speech and children's vocabulary, then the relation between input and outcome could be caused, in whole or in part, by the genes passed from parent to child. While contributions from shared genes and parent input to language development are not mutually exclusive, it remains unclear whether the size and significance of input effects found in previous studies would persist in the absence of genetic confounds. It is also possible that without these confounds, differences in diet, education,

Abbreviations: CAP, Colorado Adoption Project; CDI:WS, Communicative Development Inventory: Words and Sentences; MLU, mean length of utterance; SES, socioeconomic status; VIF, variance inflation factor.

or other factors could instead account for environmental variation in language ability (see Stromswold, 2001 for review). We will refer to the alternative as the *genetic counterhypothesis*.

There has been very little work to date that directly addresses the *genetic counterhypothesis*. The studies of parental input, listed above, all used samples where it is safe to assume that most or all the children shared a genetic relationship with their caregiver (but see Gauthier & Genesee, 2011; Hardy-Brown et al., 1981; Huttenlocher et al., 2002 discussed below). The *input hypothesis* has been central to public policy (e.g., Wong et al., 2018), the design of interventions (e.g., Suskind et al., 2016; Weber et al., 2017), and our understanding of cultural variability (e.g., Mastin & Vogt, 2016; Shneidman & Goldin-Meadow, 2012). Before this hypothesis can be accepted, it is critical to rule out the possibility that the input-outcome correlations are solely attributable to shared genes. The present experiment seeks to do this by investigating the relation between parental input and vocabulary acquisition in infants and preschoolers who have been internationally adopted into English-speaking homes. The remainder of the introduction motivates the current study by discussing: (1) the evidence for a genetic contribution to language development from twin studies and adoption studies, and (2) the handful of prior input studies that address the *genetic counterhypothesis*, specifically studies of non-parent caregivers.

Behavioral genetics

Twin studies

The field of behavioral genetics provides compelling reasons to seriously consider the *genetic counterhypothesis*. The hypothesis rests on two assumptions: (1) there is heritable variation in language skills which can be passed down from parent to child, and (2) variation in these skills produces variation in the input children hear. The first assumption has been extensively studied using the twin study design. Like most human traits, linguistic skill and vocabulary knowledge are more highly correlated in monozygotic ("identical") twins than in dizygotic ("fraternal") twins. Because identical twins share the same genes, whereas fraternal twins only share half of theirs, researchers can estimate the contribution of heritable variation by comparing the magnitude of the two correlations (for review, see McGuire, 2003). Studies of typically developing adults and adolescents have consistently found 50% or more of the variation in language skills is heritable, whereas less than 10% can be linked to the shared environment of the two twins (see e.g., Bratko, 1996; Rowe et al., 1999; van den Berg et al., 2004; see also Pinker, 2003). This is broadly consistent with the findings of twin studies in general. A meta-analysis of twin studies found that 69% of studies surveyed,

encompassing nearly 18,000 traits, were consistent with a model in which no variation was attributable to shared environment. Although studies of younger children generally find larger effects of shared environment and smaller effects of genes, studies of typically developing twin pairs between the ages of one and four have consistently found genetic endowment to account for 25%–50% of variance in vocabulary, grammar, and articulation (see Hayiou-Thomas et al., 2012; Stromswold, 2001 for reviews). Thus, we should expect more of the variation that we observe in children's language abilities in input studies is attributable to shared genes than to their early environment.

The second assumption, that heritable variation in language skills shapes language input, has not been extensively studied. Parents who themselves have genes promoting language ability could speak to their children more or produce better quality input, resulting in a passive gene–environment correlation. If parent input was found to be more correlated in identical twin children than in fraternal twin children, this would suggest that variation in parent input is heritable. We do not know of any studies that assess the heritability of speech habits adult twins employ with their own children. We do know, however, that differences in vocabulary in childhood are remarkably stable into adulthood (e.g., Rice & Hoffman, 2015). There is also evidence that speech habits such as talkativeness are heritable (DeThorne & Hart, 2009; DeThorne et al., 2008). There is even evidence that twins experience variation in input from their parents that can be attributed to their genes. Dale et al. (2015) surveyed parents of children enrolled in the Twins Early Development Study about how often they engaged in "informal stimulation," or practices found to be beneficial to children's language development between the ages of 3 and 4.5 years. These practices included talking, reading, and singing. They found this input measure to be moderately heritable, with children's genes accounting for about 30% of the variance in parent stimulation at both ages. In addition, between 25% and 75% of the variance in the correlation between earlier parent stimulation and later child vocabulary could be attributed to children's genes. Thus, it is probable that input-outcome correlations reflect passive effect of parents' genes on child's genes. In sum, the findings of these twin studies demonstrate the plausibility of the *genetic counterhypothesis*.

Adoption studies

Additional evidence for the heritability of linguistic skills comes from studies comparing adopted children to their adoptive and biological parents and siblings. These studies, like twin studies, provide insight into both genetic and environmental contributions to human variation. If an ability is highly heritable, we might expect that the performance of adopted children will be

highly correlated with that of their biological parents and siblings, but not their adoptive parents. If an ability is closely shaped by the home environment, then we should expect the correlations with biological relatives to be low, whereas the correlation with adoptive siblings and relevant features of the home environment should be high (for review, see also McGuire, 2003).

The most extensive data set on the cognitive abilities of adoptive children and their biological and adoptive families comes from the Colorado Adoption Project (CAP), a longitudinal study of hundreds of adoptive families and controls, spanning more than 40 years, with measures of cognitive performance from the birth parents, the adoptive parents, target children, and their adoptive siblings (Plomin & Defries, 1983; Rhea et al., 2013). Measures of verbal ability in this study included Peabody Picture Vocabulary Test vocabulary, verbal fluency, and the verbal subscales of the Wechsler Primary and Preschool Scales of Intelligence, as well as reading measures in the older children. The findings were consistent across measures and analyses: there is heritable variation in verbal abilities that appears to increase with age, and much of this heritable variation is shared across cognitive domains, such as verbal and spatial reasoning (see Felsenfeld & Plomin, 1997; Rice et al., 1986, 1989; Wadsworth et al., 1995, 2002). In contrast, there is little evidence for an effect of early environment, including parental input, on verbal ability: the correlations between adoptive parents and their children were small and the best fitting models typically lacked this pathway (see Rice et al., 1989). Similarly, the correlations between the adopted children and the siblings that they were raised with were often small, producing models which did not include shared environment effects (see Wadsworth et al., 2002). Thus, the findings from adoption are consistent with those of the twin studies, giving additional credence to the *genetic counterhypothesis*.

Breaking the confound

Although twin and adoption studies suggest that there are sizable genetic contributions to language, they are unable to determine whether the properties of caregiver speech that predict language outcomes in input studies remain significant in the absence of shared genes. Despite the plausibility of the *genetic counterhypothesis*, few input studies acknowledge this possibility and even fewer use designs that address it. However, some studies do have bearing on this question.

Studies of non-related caregivers

One way of testing the *genetic counterhypothesis* is to examine whether input from non-related caregivers, who share no genes with their children, contributes to their

language outcomes. Because non-related caregivers share no genes with their children, the results of these studies are genetically non-confounded and contain no possibility of passive gene–environment correlation. While there have been a couple studies of this sort, the results of these studies are mixed. Huttenlocher et al. (2002) compared the comprehension of simple and multi-clausal sentences in preschool-aged children to the complexity of input provided by their parents at home, as well as by their teachers during class. They found that both parents' and teachers' proportion of complex sentences predicted children's performance on the assessment, providing evidence that even input from non-related caregivers influences children's language outcomes. However, this study only considered a relatively narrow relation between input and linguistic growth, namely the comprehension of multi-clausal utterances and the availability of such utterances in everyday situations. This choice contrasts with many of the studies in the input literature, which examine the contribution of multiple forms of input (e.g., lexical diversity, syntactic complexity, word token frequency) to language development. This study of preschool input also focuses on later aspects of language acquisition than many of the toddler input studies, as well as on communicative contexts that are quite different from interactions in the family home. Thus, it remains unclear whether these effects will generalize to earlier stages of acquisition and interactions within the home.

Studies of adopted children also allow us to measure non-confounded environmental effects. Hardy-Brown et al. (1981) observed fifty 12-month-old mother-child dyads drawn from the CAP during 15-min feeding and play sessions. In this study, parent input was measured directly from these sessions. Maternal vocalizations were coded for linguistic characteristics, such as words in isolation, imperatives, declaratives, mean-length of utterances, wh-questions, and yes-no questions, as well as non-linguistic characteristics, such as imitation of child vocalizations and the mother's contingent verbal responsiveness to infant vocalizations. Using principal components analysis, the authors derived a measure of each child's communicative competence using their scores on the verbal section of the Bayley Scales of Infant Development, characteristics of their speech in recordings, and a diary of productive speech mothers were asked to keep. No correlations between this score and any of the linguistic characteristics of the input were found, but significant correlations were found with imitation of infant vocalizations and with contingent responsiveness, suggesting that children benefit from greater engagement from non-related parents, but not from a greater amount or diversity of speech.

This study, however, has several limitations that may decrease its sensitivity to input-acquisition relations. First, a follow-up study of 50 biologically related mother-child dyads not only failed to find a correlation between any of the linguistic characteristics of input

and communicative competence except for questions, but also failed to replicate the effects of imitation and contingent responsiveness found in the adopted sample (Hardy-Brown & Plomin, 1985). Because the relation between parent input and child language has been well-established, the absence of these effects in the follow-up study undercuts the validity of the original results. Second, a single 15-min play session may not have been long enough to capture stable differences in parental speech across subjects. Furthermore, measures of individual differences in language may be less stable at 12 months than they are a few months later. Children at this age are just beginning to produce words, and consequently there is less variation in the aspects of language that are easy to measure (e.g., total words produced; Fenson et al., 1994). Tests of language and cognition in infancy also have been found to have only moderate retest reliability (Horner, 1980). The data from many of the input studies discussed above were collected across multiple sessions per subject at different points of development. As a result, it is difficult to compare these findings to those of the larger literature.

A more recent study of non-related caregivers compared vocabulary development in ten 15-month-old children adopted between the ages of 9–13 months from China to 11 French-speaking controls living with their biological families in Quebec (Gauthier & Genesee, 2011). Children were matched on the basis of age, sex, and household socioeconomic status (SES). This study measured maternal language use (e.g., utterances, types, tokens) and attention regulation strategies (e.g., joint attention, redirecting vs. following attention) during a 30-min free play session in the home taking place at 15 months. Children's vocabulary was measured using a French version of the MacArthur Bates Communicative Development Inventory at 15 and 20 months. They found that the total number of utterances mothers of adopted children used at 15 months predicted children's expressive vocabulary score reported at the same session. In addition, mothers' use of successful redirecting utterances at 15 months predicted their children's receptive vocabulary at 20 months.

However, the conclusions we can draw from this study are also limited. Although positive correlations were found between maternal language use and vocabulary development in the adopted group, there were no correlations between any of these variables reported for control children. Thus, it remains unclear whether total maternal utterances or use of redirecting utterances predicted vocabulary in non-adopted children as well. As with the CAP studies, the failure to find a link between maternal speech and vocabulary in the non-adopted control group casts doubt on the validity of these results. The study also reported no significant correlations between age at adoption and vocabulary or total number of months exposed to French and vocabulary at either 15 or 20 months, which we would otherwise expect. These findings suggest that

the number of children considered may not have been sufficient to find effects. The relatively small range of ages considered (all children were adopted between the ages of 9–13 months) might also have contributed to these problems. In addition, no corrections for multiple comparisons were run, in spite of the numerous non-significant correlations between input and vocabulary reported.

Current studies

Our studies follow two samples of children adopted from China and Eastern Europe: one sample adopted in infancy, and another sample adopted at preschool age. Internationally adopted children provide an ideal test case for a genetically non-confounded input study. Because adopted children are not biologically related to their caregivers, any relation between parent speech and vocabulary growth will not be impacted by passive gene–environment correlation. If the *input hypothesis* holds, we expect to find a positive relation between our measures of maternal speech and vocabulary growth. However, the absence of a relation between these measures would support the *genetic counterhypothesis*, suggesting that previously observed effects of maternal speech are largely attributable to genetic confounds. In addition, because our samples were adopted and observed at different ages, we can also examine whether there are age-related differences in the effects of input on vocabulary growth. The twin studies reviewed previously suggest that the effects of shared environment diminish with maturity as genetic factors play an increasing role in language outcomes. Thus, we might expect the effect size of this relation to be larger in our younger infant sample than in our older preschool sample.

Although all these children have experience with a language other than English, and might be considered bilingual, none of them continued receiving native-language exposure following adoption. Young children have been found to rapidly lose proficiency in their native language following adoption, suggesting that as long as children are not receiving input in their native language, its effect on their uptake of English is likely to be negligible (Krakow et al., 2005; Ventureyra et al., 2004). The current study will investigate whether properties of maternal speech still exert influence on vocabulary outcomes in these groups, and whether this influence is mediated by the age of the child.

STUDY 1

Methods

Participants

Twenty-nine preschool age children (2;7–6;1, $M = 4;3$) who were adopted from China ($n = 21$) and Eastern Europe

($n = 8$) were included in the study. The preschool children (21 girls and 8 boys) had been adopted between the ages of 2;5 and 5;6 ($M = 3;1$) and had been in the United States for 0.5–6 months ($M = 3.4$ months). This sample was drawn from data collected for previous studies of language acquisition in internationally adopted children (Snedeker et al., 2007, 2012).

Prior to adoption, the children in this sample are all believed to have been exposed to Russian, Mandarin, or Cantonese. Adoptive parents were contacted by posting on internet and community-based groups who were waiting to adopt children or who had recently adopted. Other families heard about the study from families who had participated. We excluded children from the analysis if they were known to have either (1) a major developmental disorder, such as Down syndrome or autism spectrum disorder, or (2) a sensorimotor impairment that might have affected language learning, such as hearing loss or a cleft palate. Children were also excluded if they continued to be regularly exposed to a non-English language by their adoptive parents. From this data, children with high-quality video recordings and vocabulary data available before they had been in the United States for 6 months were included. All children were adopted by monolingual English-speaking families and were typically developing. The mothers in the preschool group had either professional or graduate degrees ($n = 17$), college degrees ($n = 9$), or had some college ($n = 3$).

Materials and procedure

All the families in this sample participated in studies in which both language samples and vocabulary data were collected. Families recorded a language sample (i.e., a play session) with their children using a standard box of toys sent to the families. Play sessions with preschool children were included until the children had been in the United States for 6 months. Families also completed the MacArthur-Bates Communicative Development Inventory: Words and Sentences form (CDI:WS; Fenson et al., 1994) within the same month as the recording. The use of CDI:WS in this population was validated in a prior study (Snedeker et al., 2012). All sessions in this sample were recorded with mothers.

The number of observations provided for each child varied depending on (1) the study that the family had signed up for (Snedeker et al., 2012 collected monthly samples, whereas Snedeker et al., 2007 collected samples every 3 months), (2) when the family first entered the study, and (3) how long the family choose to continue participating. In previous studies, internationally adopted preschoolers were found to make rapid progress in learning English, with many children beginning to reach the ceiling of the CDI:WS within

6–9 months of arriving in the United States. For this reason, samples were limited to sessions that took place within the child's first 6 months in the United States. This resulted in 54 sessions included in the final analysis, with each of the 29 preschoolers contributing between 1 and 6 samples each ($M = 1.86$ sessions, avg. length = 29 min).

Coding

Consistent with prior studies of caregiver input (e.g., Pancsofar & Vernon-Feagans, 2006), language samples were coded for four speech measures: (1) the mean number of word tokens spoken per minute (a measure of quantity of speech provided); (2) the mean number of word types spoken per minute (a measure of lexical diversity); (3) the mean length of utterance, or MLU (a measure of the morphemic length and complexity of parents' speech); and (4) the proportion of utterances that were questions (a measure of the interactional style of the parent). Questions included yes-no questions (e.g., *Is that your crayon?*), wh-questions (e.g., *What color is that?*), and alternative questions (e.g., *Do you want to play with the truck or the car?*). These measures were also examined in a previous study of adopted children (Hardy-Brown et al., 1981). The sessions were transcribed and analyzed using CLAN (MacWhinney, 2000). Measures of vocabulary and input for each individual child can be found in Supporting Information (see Tables S1 and S2).

Analysis

The present study is a secondary analysis of a data set that was collected with another purpose in mind. However, the primary hypothesis that guided this research was a priori: based on many decades of earlier work, we expected that if input shaped vocabulary development in the absence of a genetic confound this would result in a positive correlation between the number of word types a child heard and the number of word types that they produced. Other analyses were more exploratory: We had no clear expectation at the beginning of this research about how these effects might vary across age groups. An initial analysis for this data set used single-level regression models that were fully specified before seeing the data patterns (Shafto et al., 2010). These analyses produced findings closely parallel to the present paper. However, they were inappropriate to the structure of the data set. The present multi-level models were constructed to address that limitation. Zero-order correlations between our input and vocabulary measures at last session are provided in Supporting Information (see Table S3).

Results

Vocabulary residual score

To determine the relation between input and vocabulary development in internationally adopted preschoolers, it was necessary to calculate the degree to which the child's CDI:WS vocabulary was greater than or less than one would expect given their age and the length of their exposure to English. This was estimated this by taking a larger sample of internationally adopted preschoolers ($n = 182$), who had provided CDI:WS data for earlier studies (Snedeker et al., 2007, 2012) and conducting a stepwise regression of the vocabulary score with Time in the United States ($R^2 = .54, p < .001$) and Age of Arrival ($R^2 = .03, p < .001$) as predictors. A standardized residual score was then calculated for each of the 54 sessions included in the primary analysis. This vocabulary residual is a measure of the difference between a subject's predicted vocabulary, as would be estimated by their age of arrival and time in the United States, and their reported vocabulary at that session. Positive residuals indicate a higher reported vocabulary than would be predicted, and negative residuals indicate a lower vocabulary. These residual scores are parallel to standard scores that are often used in studies in which children differ in age. By controlling for variability related to the maturity of the child at the point when acquisition begins and the amount of time that they have had to learn the language, we can focus on variation that could plausibly be related to individual differences like the quality and quantity of caregiver speech. This score served as the response variable for the subsequent analyses, which were conducted in R (R Core Team, 2017).

Stepwise regression

Because multiple observations were available for many of our participants, our data set could not be analyzed using a single-level regression model. Instead, we constructed mixed effects hierarchical linear models using the R package *lme4* (Bates et al., 2015). In these models, subject was treated as a random intercept with each session serving as a repeated measure for that subject, reflecting the underlying structure of our data set. The dependent variable in these models was the vocabulary residual of the child, and the four measures of caregiver input served as predictors. To determine the best fitting model, both a forward- and backward-selection stepwise regression analysis were conducted, using likelihood-ratio testing to assess model fit. In the forward regression, the initial base model consisted only of a random intercept by subject, then all four of the possible one-factor models (word tokens, word types, MLU, or questions) were constructed, and compared to the base model. The model with the best fit determined

by log-likelihood was selected, and the two-factor models were constructed by adding each of the remaining predictors individually to the best one-factor model. This process was repeated until no additional variables improved fit, as determined by χ^2 test, and the final model was selected. In the backward regression, the initial model included all four of the input variables. These variables were removed, one by one, until all the remaining variables were significant predictors of the response variable. Variables that had the least impact on model fit were removed first, using the same criteria as in the forward regression.

Findings

We began our forward regression with a comparison between the baseline model and each of the one-factor models (see Table 1). We found that *Model 1d*, containing MLU as a predictor of vocabulary (unstandardized: $B = .87, SE = .18$; standardized: $\beta = .50, SE = .10; p < .001$) produced the best fit compared to the baseline model ($\chi^2(1) = 18.88, p < .001$). Although *Model 1a*, containing word tokens, and *Model 1b*, containing word types, also improved fit over baseline, *Model 1d* provided the best fit of the three. Therefore, we used *Model 1d* to compare the fits of our two-factor models. We found that *Model 2c* consisting of MLU (unstandardized: $B = .632, SE = .204$; standardized: $\beta = .36, SE = .11; p = .001$) and word types per minute (unstandardized: $B = .102, SE = .038$; standardized: $\beta = .28, SE = .10; p = .007$) provided the greatest improvement of fit over *Model 1d* ($\chi^2(1) = 5.96, p = .014$). None of the other two-factor models represented a significant improvement over *Model 1d*. Likewise, none of the three-factor models produced a better fit than *Model 2c*. Thus, *Model 2c* was the best-fitting model in the forward regression, where word types per minute (see Figure 1) and MLU (see Figure 2) were positively related to vocabulary outcomes.

In our backward regression, we compared our fully specified model containing all four predictors to each of the possible three-factor models (see Table 2). We found that *Model R3d*, which removed total question proportion, produced the best fit compared to the full model. None of the other models produced improved fits that reached significance. Therefore, we used *Model R3d* to compare the fits of our two-factor models. Of our two-factor models, only *Model R2c*, which removed word tokens, produced a better fit that reached significance. This model contained MLU and word types per minute and was identical to the best-fitting model reached by our forward regression, *Model 2c*.

Because many of our maternal speech variables were found to be correlated with each other, it is possible that multicollinearity in our model might have inflated the standard errors of our predictors. This problem could result in significant variables appearing non-significant

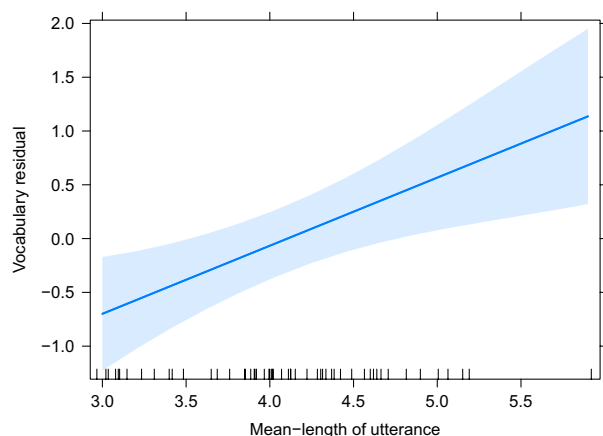
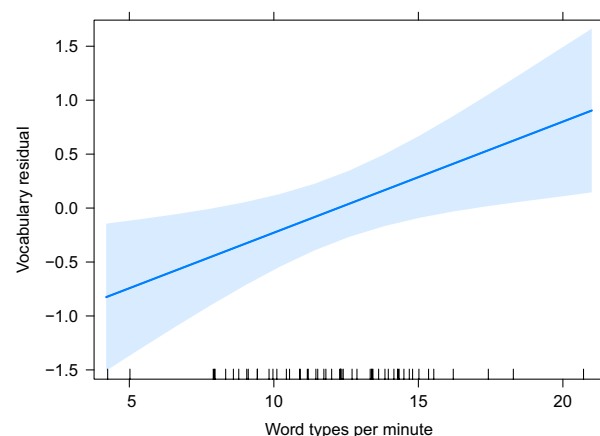
TABLE 1 Forward stepwise mixed effects regression models predicting adopted preschoolers' vocabulary from maternal speech

	Baseline	1-Factor			2-Factor			Model 2c	3-Factor	
	Base Model	Model 1a	Model 1b	Model 1c	Model 1d	Model 2a	Model 2b		Model 3a	Model 3b
Intercept (Std. Err.)	-0.047 (0.19)	-1.23 (0.53)	-2.06 (0.46)	-0.45 (0.54)	-3.57 (0.76)	-3.75 (0.81)	-3.50 (0.82)	-3.83 (0.75)	-3.46 (0.77)	-3.99 (0.81)
Word Tokens per Minute		0.019* (0.008)				0.005 (0.007)			-0.018* (0.011)	
Word Types per Minute			0.16*** (0.036)					0.10** (0.038)	0.17** (0.060)	0.10** (0.039)
Total Question Proportion				1.08 (1.35)			-0.28 (1.19)			0.67 (1.10)
Mean Length of Utterance					0.87*** (0.18)	0.83*** (0.19)	0.88*** (0.18)	0.63** (0.20)	0.61** (0.19)	0.59* (0.21)
AIC	147.90	145.47	133.43	149.24	131.02	132.54	132.98	127.05	126.38	128.72
BIC	153.87	153.42	141.39	157.19	138.97	142.48	142.92	137.00	138.31	140.66
Improved Fit? ANOVA $\chi^2(1)$		1a>Base 4.43*	1b>Base 16.47***	No 0.66	1d>Base 18.88***	No 0.48	No 0.04	2c>1d 5.96*	No 2.67	No 0.32

Note: Best fit at each step bolded. Best fit overall boxed.

Abbreviations: AIC, Akaike information criterion; BIC, Bayesian information criterion.

* $p < .1$; ** $p < .05$; *** $p < .01$; **** $p < .001$.

**FIGURE 1** Fixed effects of MLU on vocabulary in the best-fitting preschool model (Model 2c/R2c)**FIGURE 2** Fixed effects of word types per minute on vocabulary in the best-fitting preschool model (Model 2c/R2c)

and being erroneously removed from the model. To test this, we used the *car* package in R (Fox & Weisberg, 2018) to calculate variance inflation factors (VIF) for each of our predictor variables in our fully specified *Model R4*. While the VIF values suggested some multicollinearity between words tokens per minute (2.80) and word types per minute (3.16), these were below the standard problematic range of 5–10 (Stine, 1995).

Thus, our forward and backward regressions converged on the same model, one in which both parental

MLU and lexical diversity (word types per minute) predicted child vocabulary.

Post-hoc analysis

An alternative explanation of our results is that they indicate a child-to-parent effect of vocabulary on input, rather than a parent-to-child effect as predicted by the input hypothesis. Our study is limited in that samples of

TABLE 2 Backward stepwise mixed effects regression models predicting adopted preschoolers' vocabulary from maternal speech

	Full Model R4	3-Factor Model R3a	Model R3b	Model R3c	Model R3d	2-Factor Model R2a	Model R2b	Model R2c
Intercept (Std. Error)	-3.49 (0.84)	-3.99 (0.81)	-3.76 (0.89)	-2.23 (0.66)	-3.46 (0.77)	-1.69 (0.49)	-3.75 (0.81)	-3.83 (0.75)
Word Tokens per Minute	-0.018 (0.011)		0.005 (0.007)	-0.015 (0.66)	-0.018* (0.011)	-0.019 (0.011)	0.005 (0.007)	
Word Types per Minute	0.17** (0.061)	0.10** (0.039)		0.22*** (0.060)	0.17** (0.060)	0.23*** (0.060)		0.10** (0.038)
Total Question Proportion	0.012 (1.12)	0.67 (1.10)	0.037 (1.20)	1.16 (1.13)				
Mean Length of Utterance	0.61** (0.20)	0.59* (0.21)	0.83*** (0.201)		0.61** (0.19)		0.83*** (0.19)	0.63** (0.20)
AIC	128.36	128.72	134.53	134.10	126.38	133.20	132.54	127.05
BIC	142.28	140.66	146.47	146.03	138.31	143.15	142.48	137.00
Improved Fit?		No	No	No	R3d>Full	No	No	R2c>R3d

Note: Best fit at each step bolded. Best fit overall boxed.

Abbreviations: AIC, Akaike information criterion; BIC, Bayesian information criterion.

* $p < .1$; ** $p < .05$; *** $p < .01$; **** $p < .001$.

input are compared to measures of vocabulary growth collected concurrently, instead of being compared to subsequent word learning at a later time. This creates the possibility that even in the absence of a passive gene–environment correlation, our effect may be confounded by an evocative gene–environment correlation, in which the child's heritable behavior elicits certain behaviors from their parents (Plomin et al., 1977; Scarr & McCartney, 1983). For example, linguistically precocious children might elicit more speech from their caregivers. To explore this hypothesis, we conducted an analysis to determine whether changes in children's linguistic abilities predicted changes in parental speech. For this analysis, we excluded all children who had only one observation on record. The eight remaining preschoolers had an average of four observations each (33 total; range 2–5). For each observation, we calculated the change in children's raw unadjusted vocabulary score and maternal MLU and word types at the subsequent observation. We then constructed hierarchical linear mixed-effects models with mothers' change in input as the response variable and children's change in raw vocabulary as a covariate. Each model included a random intercept for subject. We then compared the log-likelihood of each model to determine whether adding change in vocabulary provided a better fit over the null model. We found no significant effect of vocabulary growth on changes in maternal MLU or maternal lexical diversity.

These results suggest that the effects of input are not due to an evocative gene–environment correlation but instead reflect effects of parental speech on children's vocabulary acquisition. Prior research has produced mixed findings on the presence of evocative gene–environment correlations in early language input. DeThorne and Hart (2009) compared conversations between twins and non-related examiners and found no evidence that the adult's speech was more similar for pairs of identical twins than for pairs of fraternal twins. In contrast, Dale et al. (2015) found evidence that parent speech to identical twin pairs was more correlated than speech to fraternal twins. The present study finds no evidence of child-to-parent effects in non-related caregivers. The differences across these studies could be attributable to differences in populations or measures.

Summary

We found that preschool-aged children who heard longer utterances and a greater variety of words had larger vocabularies than would be expected given exposure and time in the United States. The number of word tokens per minute, the raw quantity of speech, was also reliably correlated with the vocabulary residual but this effect disappeared when MLU or word tokens per minute were included in the model. This pattern suggests that

the amount of speech only predicted vocabulary to the extent the greater quantity resulted in a greater variety of words or more complex syntactic structures. Similar findings have been reported from numerous other input studies, where either MLU, word types, or both were found to be correlated with vocabulary (e.g., Hart & Risley, 1995; Hoff, 2003; Huttenlocher et al., 1991, 2010; Newman et al., 2016; Rowe, 2012; Shneidman et al., 2013; Shneidman & Goldin-Meadow, 2012).

In our next study, we examine the effects of maternal input on vocabulary in our adopted infant sample. Given our preschool findings, as well as the larger estimated effect of environmental factors on lexical development in younger children relative to older children, we expect to find similar effects of maternal speech on infants' vocabulary.

STUDY 2

Methods

Participants

Seventeen infants (1;3–2;8, $M = 2;0$) who were adopted from China ($n = 17$) were included in this sample. These children (all girls) had been recruited as a comparison group for a study of internationally adopted preschoolers (Snedeker et al., 2012). Adoptive parents of infants were recruited in the same way as the parents of preschoolers, through a combination of outreach to adoption groups and word-of-mouth. We applied the same exclusionary criteria as with the preschoolers. Infants with high-quality recordings and vocabulary data under 2;8 were included. The infants were adopted between the ages of 0;8–1;4 ($M = 1;0$) and had been in the United States longer than the preschool children (7–20 months, $M = 15$ months). All infant participants were born in China were believed to have been exposed to Mandarin or Cantonese prior to adoption. They were adopted by monolingual English-speaking families and were typically developing. Like the preschool sample, mothers in the infant group had professional or graduate degrees ($n = 11$) or had earned a college degree ($n = 6$).

Materials and procedure

The materials and procedures were the same for infants as for the preschoolers. In previous studies of this sample, infants did not begin to reach ceiling performance on the CDI:WS until 2;8 on average, slightly older than the normal age range testing using the CDI:WS (Snedeker et al., 2007, 2012). As a result, infants' play sessions were included until this age. In total, 71 sessions were collected. Three of the infants recorded some sessions with their fathers. To reduce the amount of variance in our models,

we limited our analysis to dyadic interaction between mother and child. This resulted in seven sessions being excluded from the analysis. These criteria produced 64 sessions, with each of the 17 infants contributing between 1 and 12 samples each due to differences in sampling and parent participation across studies ($M = 4.17$, avg. length = 41 min).

Coding

Infant language samples were coded for the same measures as the preschoolers' samples. Measures for individual children can be found in Supporting Information (see Tables S4 and S5).

Analysis

Our hypotheses for the present analysis were the same as for the preschool study.

Results

Vocabulary residual score

As with the preschool group, a vocabulary residual score was calculated from a larger study of internationally adopted infants ($n = 223$) using Time in the United States ($R^2 = .45$, $p < .001$) and Age of Arrival ($R^2 = .03$, $p < .001$) as covariates.

Stepwise regression

The same stepwise model comparison used to analyze the preschooler data was also used with the infants, using likelihood-ratio testing to determine variable inclusion and removal. Zero-order correlations can also be found in Supporting Information (see Table S6).

Findings

As in the preschool study, we begin our forward regression with a comparison between the baseline model and each of the one-factor models (see Table 3). We found that *Model 1d*, containing MLU (unstandardized: $B = .45$, $SE = .16$; standardized: $\beta = .39$, $SE = .14$; $p = .006$) produced the best fit compared to the baseline model ($\chi^2(1) = 7.30$, $p = .006$). While *Model 1a*, containing word tokens, was also found to produce improved fit over baseline, *Model 1d* produced a better fit. As a result, we used *Model 1d* to compare the fits of our two-factor models. However, none of the two-factor models produced a significantly better fit over the one-factor model. Thus,

	Baseline Base Model	1-Factor Model 1a	Model 1b	Model 1c	Model 1d	2-Factor Model 2a	Model 2b	Model 2c
Intercept (Std. Err.)	-0.61 (0.26)	-0.109 (0.43)	-0.027 (0.44)	-0.12 (0.61)	-1.24 (0.72)	-1.21 (0.72)	-1.24 (0.72)	-1.54 (0.81)
Word Tokens per Minute		0.011* (0.005)				0.003 (0.006)		
Word Types per Minute			0.053* (0.031)				0.008 (0.037)	
Total Question Proportion				2.13 (1.59)				1.22 (1.57)
Mean Length of Utterance					0.45** (0.16)	0.39* (0.205)	0.42* (0.202)	0.42* (0.17)
AIC	139.47	137.48	138.54	139.69	134.16	135.84	136.09	135.53
BIC	145.95	146.12	147.18	148.32	142.80	146.64	146.89	146.32
Improved Fit? ANOVA $\chi^2(1)$		1a>Base 3.98*	No 2.92*	No 1.78	1d>Base 7.30**	No 0.32	No 0.067	No 0.63

Note: Best fit at each step bolded. Best fit overall boxed.

Abbreviations: AIC, Akaike information criterion; BIC, Bayesian information criterion.

* $p < .1$; ** $p < .05$; *** $p < .01$; **** $p < .001$.

Model 1d was found to be the best-fitting model in the forward regression.

In our backward regression, we again compared our fully specified model containing all four predictors to each of the possible three-factor models (see Table 4). We found that removing any one of the four predictors from the full model improved fit. However, *Model R3d*, which removed word types per minute, produced the best fit of the four. Using *Model R3d* as our comparison case, we found that removing any one of the remaining three predictors also improved model fit. We used the best-fitting *Model R2c*, which removed word tokens and left only MLU and total question proportion, for subsequent comparisons. Finally, of the one-factor models, only removing total question proportion in *Model R1b* significantly improved model fit. This model only contained MLU and was identical to the best-fitting model reached in our forward regression, *Model 1d*. As with our preschool model, we calculated variation inflation factors for each of our predictor variables in *Model R4* to check for multicollinearity. While some evidence for multicollinearity was found for words tokens per minute (3.05) and word types per minute (3.01), both VIF values were within acceptable ranges (Stine, 1995).

Thus, our forward and backward regressions were found to converge on a single best-fitting model, where MLU was positively related to vocabulary growth (see Figure 3).

Post-hoc analysis

We also performed the same post hoc analysis to check for a possible evocative gene–environment correlation in our design. We asked whether change in infant vocabulary predicted mothers' change in MLU over time. After

excluding subjects with only one observation, we were left with five infants with an average of 10 observations per child (52 total; range 9–12). After constructing a hierarchical linear mixed-effects model with mothers' change in input as the response variable and children's change in raw vocabulary as a covariate, we found no significant effects of vocabulary growth on changes in maternal speech in either model. These results suggest that, much like the preschoolers, the observed effects of input are not due to evocative gene–environment correlation.

Summary

As in the preschooler analysis, infants who heard longer utterances on average had vocabularies higher than would be expected by age. Similarly, word tokens per minute was also significantly correlated with the vocabulary residual, but both the resulting model and the model with word tokens per minute and MLU provided a worse fit than the model with MLU alone. Unlike in our findings for preschool children, word types per minute was not found to predict infant vocabulary. None of the multi-factor models in either the forward or backward regression produced a significantly better fit than the one-factor model.

DISCUSSION

The present studies show that the relation between parental child-directed speech and vocabulary is robust in the absence of a shared genetic relationship between parent and child. In particular, the mean-length of maternal utterance was found to be positively correlated with vocabulary in both the infant and preschool groups. In

TABLE 3 Forward stepwise mixed effects regression models predicting adopted infants' vocabulary from maternal speech

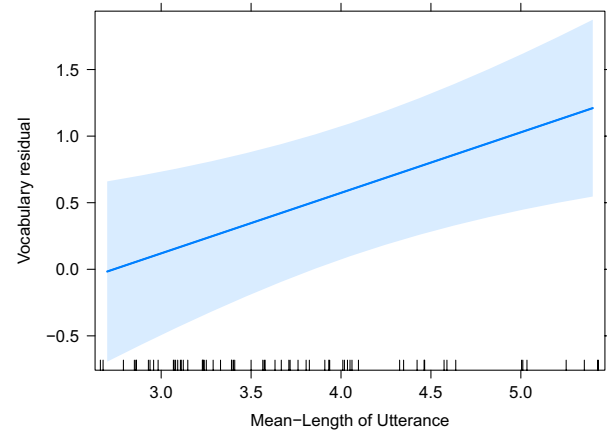
TABLE 4 Backward stepwise mixed effects regression models predicting adopted infants' vocabulary from maternal speech

Full	3-Factor			2-Factor			1-Factor			
	Model R4	Model R3a	Model R3b	Model R3c	Model R3d	Model R2a	Model R2b	Model R2c	Model R1a	Model R1b
Intercept (<i>SE</i>)	-1.53 (0.83)	-0.808 (0.707)	-1.21 (0.73)	-1.56 (0.82)	-1.53 (0.82)	-0.75 (0.67)	-1.21 (0.72)	-1.54 (0.81)	-0.12 (0.61)	-1.24 (0.72)
Word tokens per minute	0.005 (0.010)	0.009 (0.009)	0.005 (0.010)		0.004 (0.007)	0.011 (0.005)	0.003 (0.006)			
Word types per minute	-0.004 (0.053)	0.014 (0.053)	-0.010 (0.053)	0.014 (0.037)						
Total question proportion	1.33 (1.58)	1.99 (1.57)		1.32 (1.60)	1.33 (1.58)	1.95 (1.56)		1.22 (1.57)	2.13 (1.59)	
Mean length of utterance	0.35 (0.21)		0.39 (0.21)	0.37 (0.21)	0.34 (0.21)		0.39 (0.205)	0.42* (0.17)		0.45** (0.16)
AIC	139.09	139.78	137.80	137.37	137.09	137.87	135.84	135.53	139.69	134.16
BIC	154.20	152.73	150.76	150.32	150.05	148.67	146.64	146.32	148.32	142.80
Improved fit?		R3a>Full	R3b>Full	R3c>Full	R3d>Full	R2a>R3d	R2b>R3d	R2c>R3d	No	R1b>R2c

Note: Best fit at each step bolded. Best fit overall boxed.

Abbreviations: AIC, Akaike information criterion; BIC, Bayesian information criterion.

* $p < .05$; ** $p < .01$; *** $p < .001$.

**FIGURE 3** Fixed effects of MLU on vocabulary in the best-fitting infant model (Model 1d/R1b)

addition, preschoolers' vocabulary was also positively related to the diversity of lexical items present in maternal speech. Both these measures have been found to be predictive of child vocabulary in prior input studies with genetically related caregivers (Hart & Risley, 1995; Hoff, 2003; Huttenlocher et al., 1991, 2010; Newman et al., 2016; Rowe, 2012; Shneidman et al., 2013; Shneidman & Goldin-Meadow, 2012). The present findings go beyond these studies by demonstrating that these correlations persist in the absence of a genetic relationship. Therefore, the results of this study support the *input hypothesis*, suggesting that the pace of children's early vocabulary growth is determined in part by the quality of caregiver speech.

The remainder of this discussion explores: (1) how these findings relate to earlier input studies; (2) how to reconcile these findings with behavioral genetics research; (3) how these findings relate to prior research on international adoption; (4) why the findings for preschoolers and infants may have diverged; and (5) limitations of our current study design.

Relation to prior input studies

As noted above, the conclusions of the current study are readily compatible with those of the input literature. Like many of these studies, parents were recorded during one-on-one, semi-naturalistic interactions with their young children. These interactions were coded for features of child-directed speech that have previously been found to relate positively with vocabulary. Comparing our zero-order correlations between maternal speech and vocabulary (see Tables S3 and S6), we find that our effect sizes fall within the range of those found in a recent meta-analysis of input studies (Anderson et al., 2021). Our findings also converge with the three previous studies which found relations between children's language development and input from caregivers who were not biologically related to

them (Gauthier & Genesee, 2011; Hardy-Brown et al., 1981; Huttenlocher et al., 2002). This study differed from Huttenlocher et al. (2002) in two ways. First, their study explored grammatical complexity in preschool-aged children's speech and therefore could not speak to the role of input in early vocabulary acquisition. Second, they examined input from a teacher in a classroom and thus did not address how informal interactions in the home might affect language growth.

Hardy-Brown et al. (1981), in contrast to the present study, did not find effects of lexical diversity or MLU (or other structural factors) on children's communicative competence, though they did find an effect of mothers' use of contingent utterances. The present study differed from Hardy-Brown et al. (1981) in two ways which may have increased its sensitivity: (1) it used a larger sample of input and (2) it used a well-validated measure of early linguistic competence. One can reconcile the findings by hypothesizing that there are effects of input on communicative competence in adopted children (in both populations) but the Hardy-Brown et al. study lacked the sensitivity to find them. This analysis is supported by the absence of many of the commonly reported correlations between input and outcome in the control study that the researchers conducted with genetically related caregivers (Hardy-Brown & Plomin, 1985).

Finally, our results corroborate the results of Gauthier and Genesee (2011), who found that the total number of maternal utterances predicted French vocabulary in a sample of 15-month-olds adopted from China. While our studies used the same measure of vocabulary, we considered different measures of input in our infant groups which were found to be significant predictors (i.e., MLU in our study, utterances in their study). In addition, our sample encompassed a larger age range, and vocabulary in each age cohort was predicted by similar factors (i.e. MLU in our infant group; types and MLU in our preschool group). Our study also found a relation between vocabulary size and both age of adoption and time in the adopted country. Neither of these relations were found in Gauthier and Genesee (2011), possibly due to the relatively small age range considered.

Relation to behavioral genetics

Although it would be reasonable to assume that the effect of a single environmental factor (i.e., maternal speech) would be unobservable in a small sample, particularly in a population like this one where variation in input might be reduced, these findings are broadly consistent with the results of twin studies (Polderman et al., 2015; Stromswold, 2001). First, prior twin studies of vocabulary acquisition have found that, early in language acquisition, the effect of shared environment is larger than the effect of genes, but that these effects get

smaller over time (Stromswold, 2001). Both the groups in our study are in a similar point in acquisition as the youngest children in the twin studies, and thus we might expect that this would be the point at which we would be most likely see an impact of maternal speech on vocabulary growth. Secondly, these findings are compatible with previous studies of children raised in institutions, as environmental effects are larger in the context of deprivation (Rowe et al., 1999; Turkheimer et al., 2003), and placement in richer environments supports language development (Croft et al., 2007; Windsor et al., 2007, 2013). While the current findings demonstrate that the effects of parent input are not solely attributable to a passive gene-environment correlation, the heritability of language ability in early childhood is undeniable.

Relation to prior adoption studies

The present study also corroborates findings from previous studies of language development in institutionalized children pre- and post-adoption. Research on international adoption has largely focused on the effects of relocation on institutionalized children suffering from environmental deprivation. These studies have repeatedly found that movement from lower-input environments to higher-input environments supports language growth, in both first- and second-language acquisition (Croft et al., 2007; Windsor et al., 2007, 2013). While nothing is known about the prior language exposure of children in the current study, the results are readily compatible with the conclusions of these earlier studies: input matters, regardless of the genetic relationship between child and caregiver.

Differences between the findings for infants and preschoolers

One question raised by these findings is how to explain between-group differences in sensitivity to input. In the preschoolers, vocabulary size was predicted by the lexical diversity of input and in infants, it was not.

The likely explanation, as we noted earlier, is that preschoolers do not have to contend with the same maturational bottlenecks as infants, and as a result are able to gain more from richer input. Previous studies of internationally adopted children have demonstrated that while the sequence of language milestones is the same in infants as in preschoolers, the pace of language acquisition is not. In the earliest stages, preschoolers acquire words at roughly four times the rate of the infants (Snedeker et al., 2012). Similar age-related differences in language acquisition have also been reported in other samples of children adopted from China (Krakow et al., 2005). Thus, while infants may benefit from hearing more speech, they may not be able to learn words quickly

enough to fully benefit from the variability in lexical diversity present in this sample. In contrast, preschoolers may be learning so quickly that additional word types have greater marginal value. Future work could explore how input in these populations and at these ages predicts outcomes later in childhood (see Silvey et al., 2021).

Limitations

The current study design does not allow us to explore the relative contributions of genetic and environmental factors to vocabulary, though it raises questions about these effects that could be addressed in future work. In typical input studies, that enroll children who are being raised by their birth parents, the correlations between input and vocabulary growth presumably capture both purely environmental effects (like those in the present study) and the effects of shared genes (documented in the twin studies). Thus, it is likely that the effect size for parent input on vocabulary will be smaller in adopted children than it is in non-adopted children, because any part of the correlation that is attributable to shared genes will be absent. Future studies could compare the size of these correlations in carefully matched groups of adopted and non-adopted children. Such studies would best be done in contexts in which adoption occurs in early infancy and does not coincide with a switch in the language environment. Our study design also does not allow us to explore the role of genetic factors in the vocabulary growth of adopted children: doing so requires collecting information about the language abilities of the biological parents or siblings of the adopted children, something that is not practical in our sample.

Our study is also not designed to explore the possibility of gene–environment interactions, where environmental factors influence the magnitude of heritability estimates. One factor that has been found to moderate heritability estimates is SES. For example, Turkheimer et al. (2003) surveyed an economically diverse sample of twins drawn from the National Collaborative Perinatal Project to examine possible gene-by-environment interactions in IQ score components. They found greater environmental contributions to IQ in lower SES families as compared to higher SES families. This points to the possibility that such gene–environment interactions might exist in children's early cognitive development, possibly impacting language development. However, it is unlikely that this particular interaction would be highly relevant to this data set, given the high average education and SES of our subject families. In general, we expect gene–environment interactions to be largest in samples drawn from diverse environments, where variance in early experiences is high. The relative homogeneity of our sample families would reduce the likelihood of observing any large interactions. In fact, we would expect smaller effects of input on language development overall in the

presence of a gene–environment interaction, as failing to account for it would lead us to average across subgroups within our sample who respond differently to input.

Nevertheless, one should not expect the size of input effects to remain constant across populations and measures. There are many reasons why the effects of shared environment might differ across adopted and non-adopted samples. For example, the influence of environmental factors might be smaller in adoptees due to the screening process prospective parents undergo prior to adoption (U.S. Citizenship and Immigration Services, 2020). These requirements on income and other relevant socioeconomic measures might reduce environmental variation in adopted samples as compared to non-adopted samples. On the other hand, the influence of environmental factors might be greater in adopted children due to having lived in deprivation prior to placement (Croft et al., 2007; Windsor et al., 2007, 2013). Adopted children also contend with fewer maturational bottlenecks relative to non-adopted children. Because internationally adopted children begin learning English at an older age than their non-adopted peers, they benefit from more advanced development in areas of cognition relevant to language learning, such as working memory and inhibitory control. The current design suggests that these bottleneck effects are significant, as preschool-aged adoptees are additionally impacted by lexical diversity in input.

Although we conducted post hoc analyses on our data to address a possible evocative gene–environment correlation in our study, there are additional sources of non-genetic child-to-parent effects that our design is unable to address. A possible contribution to such an effect could come from non-random placement of children to adoptive families. In many contexts, adoptive parents or placement officials may show a bias to try to place children in families that they believe are similar to the child in some way. Talkative parents, for example, may select children on the basis of their verbal ability. This selection bias may create a genetic confound without any shared genes between parents and children. However, while families are often given some information about their child prior to international adoption, this information does not generally relate to variation within typical development and parents do not have the opportunity for an in-person meeting prior to selection (U.S. Citizenship and Immigration Services, 2020). We believe these factors, in addition to the homogeneous characteristics of adoptive families due to the strict requirements involved, make the possibility of non-random placement unlikely.

Our study is also unable to compare maternal input and vocabulary outcomes longitudinally: due to the lack of multiple observations for the majority of our subjects, we are only able to compare these measures concurrently. A longitudinal design is better at establishing a causal

relation between these measures, as input at a previous time can be argued to impact vocabulary growth from that time to a future time. In contrast, a concurrent design can only establish a correlation between input and vocabulary at a single point in time. Despite this, we believe that our design is adequate to argue for causation. For one, we take our samples to be representative of typical speech in the home prior to language assessment. In addition, our measure of vocabulary was drawn from parent reports, rather than the same speech sample from which input was drawn. As a result, there is little chance of the two being jointly affected by the circumstances of collection.

An additional limitation is our reliance on parent report measures. Because we did not assess children's vocabulary directly, these measures could reflect differences in parental perceptions in addition to true differences in children's production. While we cannot rule out this possibility, there are good reasons for thinking that it cannot account for the present findings. First, the English version of the CDI:WS underwent extensive validation during its development (see Fenson et al., 1994) and subsequent studies have documented close correlations between reported CDI:WS vocabulary and measures taken in the lab or in the home (e.g., Muhinyi & Rowe, 2019; Salo et al., 2016; Taumoepeau, 2016). Second, previous work with this data set demonstrated that CDI:WS scores in internationally adopted children are also correlated with the lexical diversity of their spontaneous language production (Snedeker et al., 2012).

It is also important to note that we have only examined a few characteristics of maternal speech that have been found to predict language growth in young children (Rowe & Snow, 2020). The previously reviewed studies of non-caregiver input found that behavioral aspects of speech, such as attention regulating utterances (Gauthier & Genesee, 2011) or verbal responsiveness (Hardy-Brown et al., 1981), also predicted children's language growth. On the basis of our current analysis, we are unable to determine whether other non-linguistic aspects of caregiver speech predict vocabulary in the absence of genetic confounds.

Finally, we are unable to make claims about the possible impact of caregiver speech on other aspects of language development, such as grammatical development. It is possible that grammatical development, which some hold to emerge separately from lexical knowledge, might be less affected by speech input, and that previously observed effects reflect a genetic confound. We find this possibility to be unlikely for the following reasons. For one, previous twin studies have found that lexical and grammatical development are not only tightly correlated, but also share much of their variance stemming from genetic and environmental factors (Dale et al., 2000). The same data set also found evidence for bidirectional effects of lexical and grammatical development (Dionne et al., 2003). As such, we find little evidence motivating

the belief that grammatical development is more genetically driven than lexical development.

CONCLUSION

In conclusion, this study found that the relation between children's vocabulary outcomes and the speech they hear at home persists even when children share no genetic relationship with their caregivers. In addition, age-related differences in the impact of parent input were also found, likely related to differences in vocabulary growth trajectories between the two age groups tested. These findings suggest a causal role of speech input on vocabulary development.

ACKNOWLEDGMENTS

We would like to thank Kailee Zhu, London Dixon, Abbie Claflin, Nicole Gavel, Ellen Godena, Candice Ishikawa, Corinne Jones, Eva Liggett, Angela Lou, John Ste Marie, Ryan Sykora, Cathy Tillman, and K. Yvonne Woodworth for assistance with data collection and transcription, as well as all the families who worked with us to make this all possible. This project was generously supported by a grant from the NSF (No. 0418423).

CONFLICT OF INTEREST

We have no known conflict of interest to disclose.

ORCID

Joseph R. Coffey  <https://orcid.org/0000-0001-6735-0644>

REFERENCES

- Anderson, N. J., Graham, S. A., Prime, H., Jenkins, J. M., & Madigan, S. (2021). Linking quality and quantity of parental linguistic input to child language skills: A meta-analysis. *Child Development, 92*(2), 484–501. <https://doi.org/10.1111/cdev.13508>
- Bates, D., Mächler, M., Bolker, B., & Walker, S. (2015). Fitting linear mixed-effects models using lme4. *Journal of Statistical Software, 67*(1), 1–48. <https://doi.org/10.18637/jss.v067.i01>
- Bratko, D. (1996). Twin study of verbal and spatial abilities. *Personality and Individual Differences, 21*(4), 621–624. [https://doi.org/10.1016/0191-8869\(96\)00091-8](https://doi.org/10.1016/0191-8869(96)00091-8)
- Brent, M. R., & Siskind, J. M. (2001). The role of exposure to isolated words in early vocabulary development. *Cognition, 81*(2), B33–B44. [https://doi.org/10.1016/S0010-0277\(01\)00122-6](https://doi.org/10.1016/S0010-0277(01)00122-6)
- Croft, C., Beckett, C., Rutter, M., Castle, J., Colvert, E., Groothues, C., Hawkins, A., Kreppner, J., Stevens, S. E., & Sonuga-Barke, E. J. (2007). Early adolescent outcomes of institutionally-deprived and non-deprived adoptees. II: Language as a protective factor and a vulnerable outcome. *Journal of Child Psychology and Psychiatry, 48*(1), 31–44. <https://doi.org/10.1111/j.1469-7610.2006.01689.x>
- Dale, P. S., Dionne, G., Eley, T. C., & Plomin, R. (2000). Lexical and grammatical development: A behavioural genetic perspective. *Journal of Child Language, 27*(3), 619–642. <https://doi.org/10.1017/S0305000900004281>
- Dale, P. S., Tosto, M. G., Hayiou-Thomas, M. E., & Plomin, R. (2015). Why does parental language input style predict child language development? A twin study of gene–environment correlation.

- Journal of Communication Disorders*, 57, 106–117. <https://doi.org/10.1016/j.jcomdis.2015.07.004>
- DeThorne, L. S., & Hart, S. A. (2009). Use of the twin design to examine evocative gene-environment effects within a conversational context. *International Journal of Developmental Science*, 3(2), 175–194. <https://doi.org/10.3233/DEV-2009-3207>
- DeThorne, L. S., Petrill, S. A., Hart, S. A., Channell, R. W., Campbell, R. J., Deater-Deckard, K., Thompson, L. A., & Vandenberg, D. J. (2008). Genetic effects on children's conversational language use. *Journal of Speech, Language, and Hearing Research*, 51(2), 423–435. [https://doi.org/10.1044/1092-4388\(2008/031\)](https://doi.org/10.1044/1092-4388(2008/031))
- Dionne, G., Dale, P. S., Boivin, M., & Plomin, R. (2003). Genetic evidence for bidirectional effects of early lexical and grammatical development. *Child Development*, 74(2), 394–412. <https://doi.org/10.1111/1467-8624.7402005>
- Felsenfeld, S., & Plomin, R. (1997). Epidemiological and offspring analyses of developmental speech disorders using data from the Colorado Adoption Project. *Journal of Speech, Language, and Hearing Research*, 40(4), 778–791. <https://doi.org/10.1044/jslhr.4004.778>
- Fenson, L., Dale, P. S., Reznick, J. S., Bates, E., Thal, D. J., Pethick, S. J., Tomasello, M., Mervis, C. B., & Stiles, J. (1994). Variability in early communicative development. *Monographs of the Society for Research in Child Development*, 59(5), v–173. <https://doi.org/10.2307/1166093>
- Fox, J., & Weisberg, S. (2018). *An R companion to applied regression*. Sage Publications.
- Gauthier, K., & Genesee, F. (2011). Language development in internationally adopted children: A special case of early second language learning. *Child Development*, 82(3), 887–901. <https://doi.org/10.1111/j.1467-8624.2011.01578.x>
- Hardy-Brown, K., & Plomin, R. (1985). Infant communicative development: Evidence from adoptive and biological families for genetic and environmental influences on rate differences. *Developmental Psychology*, 21(2), 378–385. <https://doi.org/10.1037/0012-1649.21.2.378>
- Hardy-Brown, K., Plomin, R., & DeFries, J. C. (1981). Genetic and environmental influences on the rate of communicative development in the first year of life. *Developmental Psychology*, 17(6), 704–717. <https://doi.org/10.1037/0012-1649.17.6.704>
- Hart, B., & Risley, T. R. (1995). *Meaningful differences in the everyday experience of young American children*. Paul H. Brookes Publishing.
- Hayiou-Thomas, M. E., Dale, P. S., & Plomin, R. (2012). The etiology of variation in language skills changes with development: A longitudinal twin study of language from 2 to 12 years. *Developmental Science*, 15(2), 233–249. <https://doi.org/10.1111/j.1467-7687.2011.01119.x>
- Hirsh-Pasek, K., Adamson, L. B., Bakeman, R., Owen, M. T., Golinkoff, R. M., Pace, A., Yust, P. K., & Suma, K. (2015). The contribution of early communication quality to low-income children's language success. *Psychological Science*, 26(7), 1071–1083. <https://doi.org/10.1177/0956797615581493>
- Hoareau, M., Yeung, H. H., & Nazzi, T. (2019). Infants' statistical word segmentation in an artificial language is linked to both parental speech input and reported production abilities. *Developmental Science*, 22(4), e12803. <https://doi.org/10.1111/desc.12803>
- Hoff, E. (2003). The specificity of environmental influence: Socioeconomic status affects early vocabulary development via maternal speech. *Child Development*, 74(5), 1368–1378. <https://doi.org/10.1111/1467-8624.00612>
- Horner, T. M. (1980). Test-retest and home-clinic characteristics of the Bayley Scales of Infant Development in nine- and fifteen-month-old infants. *Child Development*, 51(3), 751–758. <https://doi.org/10.1111/j.1467-8624.1980.tb02610.x>
- Hurtado, N., Marchman, V. A., & Fernald, A. (2008). Does input influence uptake? Links between maternal talk, processing speed and vocabulary size in Spanish-learning children. *Developmental Science*, 11(6), F31–F39. <https://doi.org/10.1111/j.1467-7687.2008.00768.x>
- Huttenlocher, J., Haight, W., Bryk, A., Seltzer, M., & Lyons, T. (1991). Early vocabulary growth: Relation to language input and gender. *Developmental Psychology*, 27(2), 236–248. <https://doi.org/10.1037/0012-1649.27.2.236>
- Huttenlocher, J., Vasilyeva, M., Cymerman, E., & Levine, S. (2002). Language input and child syntax. *Cognitive Psychology*, 45(3), 337–374. [https://doi.org/10.1016/S0010-0285\(02\)00500-5](https://doi.org/10.1016/S0010-0285(02)00500-5)
- Huttenlocher, J., Waterfall, H., Vasilyeva, M., Vevea, J., & Hedges, L. V. (2010). Sources of variability in children's language growth. *Cognitive Psychology*, 61(4), 343–365. <https://doi.org/10.1016/j.cogpsych.2010.08.002>
- Krakow, R. A., Tao, S., & Roberts, J. (2005). Adoption age effects on English language acquisition: Infants and toddlers from China. *Seminars in Speech and Language*, 26(1), 33–43. <https://doi.org/10.1055/s-2005-864214>
- MacWhinney, B. (2000). *The CHILDES project: Tools for analyzing talk*. Erlbaum.
- Mastin, J. D., & Vogt, P. (2016). Infant engagement and early vocabulary development: A naturalistic observation study of Mozambican infants from 1;1 to 2;1. *Journal of Child Language*, 43(2), 235–264. <https://doi.org/10.1017/S0305000915000148>
- McGuire, S. (2003). The heritability of parenting. *Parenting: Science and Practice*, 3(1), 73–94. https://doi.org/10.1207/S15327922PAR0301_04
- Muhinyi, A., & Rowe, M. L. (2019). Shared reading with preverbal infants and later language development. *Journal of Applied Developmental Psychology*, 64, 101053. <https://doi.org/10.1016/j.appdev.2019.101057>
- Newman, R. S., Rowe, M. L., & Ratner, N. B. (2016). Input and uptake at 7 months predicts toddler vocabulary: The role of child-directed speech and infant processing skills in language development. *Journal of Child Language*, 43(5), 1158–1173. <https://doi.org/10.1017/S0305000915000446>
- Newport, E., Gleitman, H., & Gleitman, L. (1977). Mother, I'd rather do it myself: Some effects and non-effects of maternal speech style. In C. E. Snow & C. A. Ferguson (Eds.), *Talking to children* (pp. 109–149). Cambridge University Press.
- Pan, B. A., Rowe, M. L., Singer, J. D., & Snow, C. E. (2005). Maternal correlates of growth in toddler vocabulary production in low-income families. *Child Development*, 76(4), 763–782. <https://doi.org/10.1111/1467-8624.00498-i1>
- Pancsofar, N., & Vernon-Feagans, L. (2006). Mother and father language input to young children: Contributions to later language development. *Journal of Applied Developmental Psychology*, 27(6), 571–587. <https://doi.org/10.1016/j.appdev.2006.08.003>
- Pinker, S. (2003). *The blank slate: The modern denial of human nature*. Penguin.
- Plomin, R., & DeFries, J. C. (1983). The Colorado Adoption Project. *Child Development*, 54(2), 276–289. <https://doi.org/10.2307/1129691>
- Plomin, R., DeFries, J. C., & Loehlin, J. C. (1977). Genotype-environment interaction and correlation in the analysis of human behavior. *Psychological Bulletin*, 84(2), 309–322. <https://doi.org/10.1037/0033-2909.84.2.309>
- Polderman, T. J., Benyamin, B., De Leeuw, C. A., Sullivan, P. F., Van Bochoven, A., Visscher, P. M., & Posthuma, D. (2015). Meta-analysis of the heritability of human traits based on fifty years of twin studies. *Nature Genetics*, 47(7), 702–709. <https://doi.org/10.1038/ng.3285>
- R Core Team. (2017). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. <https://www.R-project.org/>
- Ramírez-Esparza, N., García-Sierra, A., & Kuhl, P. K. (2014). Look who's talking: Speech style and social context in language input to infants are linked to concurrent and future speech development. *Developmental Science*, 17(6), 880–891. <https://doi.org/10.1111/desc.12172>

- Ramírez-Esparza, N., García-Sierra, A., & Kuhl, P. K. (2017). The impact of early social interactions on later language development in Spanish-English bilingual infants. *Child Development, 88*(4), 1216–1234. <https://doi.org/10.1111/cdev.12648>
- Rice, T., Carey, G., Fulker, D. W., & DeFries, J. C. (1989). Multivariate path analysis of specific cognitive abilities in the Colorado Adoption Project: Conditional path model of assortative mating. *Behavior Genetics, 19*(2), 195–207. <https://doi.org/10.1007/BF01065904>
- Rice, T., Fulker, D. W., & DeFries, J. C. (1986). Multivariate path analysis of specific cognitive abilities in the Colorado Adoption Project. *Behavior Genetics, 16*(1), 107–125. <https://doi.org/10.1007/BF01065482>
- Rice, M. L., & Hoffman, L. (2015). Predicting vocabulary growth in children with and without specific language impairment: A longitudinal study from 2; 6 to 21 years of age. *Journal of Speech, Language, and Hearing Research, 58*(2), 345–359. https://doi.org/10.1044/2015_JSLHR-L-14-0150
- Rhea, S. A., Bricker, J. B., Wadsworth, S. J., & Corley, R. P. (2013). The Colorado adoption project. *Twin Research and Human Genetics, 16*(1), 358–365. <https://doi.org/10.1017/thg.2012.109>
- Romeo, R. R., Leonard, J. A., Robinson, S. T., West, M. R., Mackey, A. P., Rowe, M. L., & Gabrieli, J. D. (2018). Beyond the 30-million-word gap: Children's conversational exposure is associated with language-related brain function. *Psychological Science, 29*(5), 700–710. <https://doi.org/10.1177/0956797617742725>
- Rowe, D. C., Jacobson, K. C., & Van den Oord, E. J. (1999). Genetic and environmental influences on vocabulary IQ: Parental education level as moderator. *Child Development, 70*(5), 1151–1162. <https://doi.org/10.1111/1467-8624.00084>
- Rowe, M. L. (2008). Child-directed speech: Relation to socioeconomic status, knowledge of child development and child vocabulary skill. *Journal of Child Language, 35*(1), 185–205. <https://doi.org/10.1017/S0305000907008343>
- Rowe, M. L. (2012). A longitudinal investigation of the role of quantity and quality of child-directed speech in vocabulary development. *Child Development, 83*(5), 1762–1774. <https://doi.org/10.1111/j.1467-8624.2012.01805.x>
- Rowe, M. L., Leech, K. A., & Cabrera, N. (2017). Going beyond input quantity: Wh-questions matter for toddlers' language and cognitive development. *Cognitive Science, 41*, 162–179. <https://doi.org/10.1111/cogs.12349>
- Rowe, M. L., & Snow, C. E. (2020). Analyzing input quality along three dimensions: Interactive, linguistic, and conceptual. *Journal of Child Language, 47*(1), 5–21. <https://doi.org/10.1017/S0305000919000655>
- Salo, V. C., Rowe, M. L., Leech, K. A., & Cabrera, N. J. (2016). Low-income fathers' speech to toddlers during book reading versus toy play. *Journal of Child Language, 43*(6), 1385–1399. <https://doi.org/10.1017/S0305000915000551>
- Scarr, S., & McCartney, K. (1983). How people make their own environments: A theory of genotype environment effects. *Child Development, 54*(2), 424–435. <https://doi.org/10.1111/j.1467-8623.1983.tb03884.x>
- Shafro, C. L., Geren, J., & Snedeker, J. (2010). *Effects of maternal input on language in the absence of genetic confounds: Vocabulary development in internationally adopted children*. In S. Ohlsson & R. Catrambone (Eds.), *CogSci 2010. Proceedings of the 32nd Annual Meeting of the Cognitive Science Society: August 11–14, 2010, Portland, Oregon*, pp. 2775–2780. Cognitive Science Society.
- Shneidman, L. A., Arroyo, M. E., Levine, S. C., & Goldin-Meadow, S. (2013). What counts as effective input for word learning? *Journal of Child Language, 40*(3), 672–686. <https://doi.org/10.1017/S0305000912000141>
- Shneidman, L. A., & Goldin-Meadow, S. (2012). Language input and acquisition in a Mayan village: How important is directed speech? *Developmental Science, 15*(5), 659–673. <https://doi.org/10.1111/j.1467-7687.2012.01168.x>
- Silvey, C., Demir-Lira, Ö. E., Goldin-Meadow, S., & Raudenbush, S. W. (2021). Effects of time-varying parent input on children's language outcomes differ for vocabulary and syntax. *Psychological Science, 32*(4), 536–548. <https://doi.org/10.1177/0956797620970559>
- Snedeker, J., Geren, J., & Shafro, C. L. (2007). Starting over: International adoption as a natural experiment in language development. *Psychological Science, 18*(1), 79–87. <https://doi.org/10.1111/j.1467-9280.2007.01852.x>
- Snedeker, J., Geren, J., & Shafro, C. L. (2012). Disentangling the effects of cognitive development and linguistic expertise: A longitudinal study of the acquisition of English in internationally-adopted children. *Cognitive Psychology, 65*(1), 39–76. <https://doi.org/10.1016/j.cogpsych.2012.01.004>
- Song, L., Spier, E. T., & Tamis-LeMonda, C. S. (2013). Reciprocal influences between maternal language and children's language and cognitive development in low-income families. *Journal of Child Language, 41*(2), 305–326. <https://doi.org/10.1017/S0305000912000700>
- Stine, R. A. (1995). Graphical interpretation of variance inflation factors. *The American Statistician, 49*(1), 53–56. <https://doi.org/10.2307/2684812>
- Stromswold, K. (2001). The heritability of language: A review and metaanalysis of twin, adoption, and linkage studies. *Language, 77*(4), 647–723. <https://doi.org/10.1353/lan.2001.0247>
- Suskind, D. L., Leffel, K. R., Graf, E., Hernandez, M. W., Gunderson, E. A., Sapolich, S. G., Suskind, E., Leininger, L., Goldin-Meadow, S., & Levine, S. C. (2016). A parent-directed language intervention for children of low socioeconomic status: A randomized controlled pilot study. *Journal of Child Language, 43*(2), 366–406. <https://doi.org/10.1017/S0305000915000033>
- Taumoepeau, M. (2016). Maternal expansions of child language relate to growth in children's vocabulary. *Language Learning and Development, 12*(4), 429–446. <https://doi.org/10.1080/15475441.2016.1158114>
- Turkheimer, E., Haley, A., Waldron, M., d'Onofrio, B., & Gottesman, I. I. (2003). Socioeconomic status modifies heritability of IQ in young children. *Psychological Science, 14*(6), 623–628. <https://doi.org/10.1046/j.0956-7976.2003.pscil475.x>
- U.S. Citizenship and Immigration Services. (2020, January 10). *Adoption* [Online resource]. <https://www.uscis.gov/>
- van den Berg, S. M., Posthuma, D., & Boomsma, D. I. (2004). A longitudinal genetic study of vocabulary knowledge in adults. *Twin Research and Human Genetics, 7*(3), 284–291. <https://doi.org/10.1375/twin.7.3.284>
- Ventureyra, V. A., Pallier, C., & Yoo, H. Y. (2004). The loss of first language phonetic perception in adopted Koreans. *Journal of Neurolinguistics, 17*(1), 79–91. [https://doi.org/10.1016/S0911-6044\(03\)00053-8](https://doi.org/10.1016/S0911-6044(03)00053-8)
- Weber, A., Fernald, A., & Diop, Y. (2017). When cultural norms discourage talking to babies: Effectiveness of a parenting program in rural Senegal. *Child Development, 88*(5), 1513–1526. <https://doi.org/10.1111/cdev.12882>
- Weisleder, A., & Fernald, A. (2013). Talking to children matters: Early language experience strengthens processing and builds vocabulary. *Psychological Science, 24*(11), 2143–2152. <https://doi.org/10.1177/0956797613488145>
- Windsor, J., Glaze, L. E., & Koga, S. F. (2007). Language acquisition with limited input: Romanian institution and foster care. *Journal of Speech, Language, and Hearing Research, 50*(5), 1365–1381. [https://doi.org/10.1044/1092-4388\(2007\)095](https://doi.org/10.1044/1092-4388(2007)095)
- Windsor, J., Moraru, A., Nelson, C. A., Fox, N. A., & Zeanah, C. H. (2013). Effect of foster care on language learning at eight years: Findings from the Bucharest Early Intervention Project. *Journal*

of *Child Language*, 40(3), 605–627. <https://doi.org/10.1017/S0305000912000177>

- Wong, K., Boben, M., & Thomas, C. (2018). *Disrupting the early learning status quo: Providence talks as an innovative policy in diverse urban communities*. PT. <http://www.providencetalks.org/wp-content/uploads/2018/07/updated-brown-eval.pdf>
- Wadsworth, S. J., Corley, R. P., Hewitt, J. K., Plomin, R., & DeFries, J. C. (2002). Parent–offspring resemblance for reading performance at 7, 12 and 16 years of age in the Colorado Adoption Project. *Journal of Child Psychology and Psychiatry*, 43(6), 769–774. <https://doi.org/10.1111/1469-7610.00085>
- Wadsworth, S. J., DeFries, J. C., Fulker, D. W., & Plomin, R. (1995). Cognitive ability and academic achievement in the Colorado Adoption Project: A multivariate genetic analysis of parent–offspring and sibling data. *Behavior Genetics*, 25(1), 1–15. <https://doi.org/10.1007/BF02197237>

SUPPORTING INFORMATION

Additional supporting information may be found in the online version of the article at the publisher's website.

How to cite this article: Coffey, J. R., Shafto, C. L., Geren, J. C., & Snedeker, J. (2022). The effects of maternal input on language in the absence of genetic confounds: Vocabulary development in internationally adopted children. *Child Development*, 93, 237–253. <https://doi.org/10.1111/cdev.13688>